① Calculati

a) $\int_{|z-2|=4} \frac{2z^3+z^2+4}{z^4+4z^2}\,dz \overset{\text{Not.}}{=\!=} I$

$z^4+4z^2=0 \Rightarrow z^2(z^2+4)=0 \Rightarrow z=0$ pol de ordinul 2 $\Rightarrow$

$z=\pm 2i$ poli de ordinul 1

$\{0;\pm 2i\} \in \text{Int}|z-2|=4$

$\overset{\text{Th. Rezid.}}{\Rightarrow} \int_{|z-2|=4} f(z)dz = 2\pi i\left(\text{Rez}(f,0)+\text{Rez}(f,2i)+\text{Rez}(f,-2i)\right)$

$\text{Rez}(f,0) = \lim_{z\to 0}\left(z^2\cdot\frac{2z^3+z^2+4}{z^4+4z^2}\right)^{(1)} = \lim_{z\to 0}\left(\frac{2z^3+z^2+4}{z^2+4}\right)'$

$= \lim_{z\to 0}\frac{(6z^2+2z)(z^2+4)-(2z^3+z^2+4)\cdot 2z}{(z^2+4)^2} = 0$

($6z^2+2z \to 0$, $2z^3+z^2+4 \to 4$, $(z^2+4)^2 \to 4$)

$\text{Rez}(f,-2i) = \lim_{z\to -2i}\left[(z+2i)\cdot\frac{2z^3+z^2+4}{z^4+4z^2}\right]$

$z^4+4z^2 = z^2(z-2i)$

$= \lim_{z\to -2i}\frac{2z^3+z^2+4}{z^2(z-2i)} = \frac{2\cdot 8i-4+4}{-4(-2i-2i)} = \frac{2\cdot 8i}{16i} = 1$

$$\mathrm{Rez}(f, 2i) = \lim_{z \to 2i} \left[(z-2i) \cdot \frac{2z^3+z^2+4}{z^4+4z^2} \right]$$

$$z^4+4z^2 = z^2(z+2i)$$

$$= \lim_{z \to 2i} \frac{2z^3+z^2+4}{z^2(z+2i)} = \frac{2(-8i)+(-4)+4}{-4 \cdot 4i} = \frac{-16i}{-16i} = 1$$

$$\gamma = 2\pi i (0+1+1) = 2\pi i \cdot 2 = 4\pi i$$

b) $$\int_{|z+i|=1,4} \frac{z^2}{z^2-1} dz \overset{\text{Not}}{=\!=} \gamma$$

$$z^2-1=0 \Rightarrow z^2=1 \Rightarrow z = \pm 1$$

pt. $z=1 \Rightarrow |z+i| = \sqrt{2} > 1,4$
$z=-1 \Rightarrow |z+i| = \sqrt{2} > 1,4$

$\Rightarrow z_{1,2} \notin \text{Int}|z+i| \Longrightarrow$ Th. Cauchy $\Longrightarrow \gamma = 0$

c) $$\int_{|z|=1} \frac{e^{2z}}{\pi z - i} dz \overset{\text{Not.}}{=\!=} \gamma$$

$$\pi z - i = 0 \Rightarrow z = \frac{i}{\pi}$$

$$\gamma = \frac{1}{\pi} \int_{|z|=1} \frac{e^{2z}}{z - \frac{i}{\pi}} dz = \frac{1}{\pi} \cdot 2\pi i f\left(\frac{i}{\pi}\right) = 2i \cdot e^{2i/\pi}$$

d) $\int_{|z|=1} \frac{ch2z}{(z-\frac{1}{2})^4} dz \overset{\text{Not.}}{=} y$

$(z-\frac{1}{2})^4 = 0 \Rightarrow z = \frac{1}{2}$ pol de ordinul 4

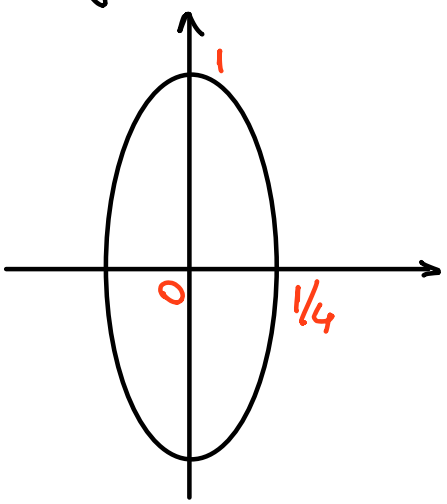
$y = 2\pi i \operatorname{Rez}(f, \frac{1}{2})$

$\operatorname{Rez}(f, \frac{1}{2}) = \frac{1}{3!} \cdot \lim_{z \to \frac{1}{2}} \left[(z-\frac{1}{2})^4 \cdot \frac{ch2z}{(z-\frac{1}{2})^4} \right]^{(3)} = \frac{1}{3!} \lim_{z \to \frac{1}{2}} (ch(2z))^{(3)}$

$= \frac{1}{3!} \lim_{z \to \frac{1}{2}} (8\, sh(2z)) = \frac{8}{6} sh1 = \frac{4}{3} sh(1)$

$y = 2\pi i \frac{4}{3} sh(1) = \frac{8\pi i}{3} sh(1)$

e) $\int_\gamma \frac{tg \pi z}{z^2} dz \overset{\text{Not.}}{=} y$ $\quad \gamma: 16x^2 + y^2 = 1$



pt $z = 0 \Rightarrow \lim_{z \to 0} \frac{tg \pi z}{z^2} =$

$= \lim_{z \to 0} \frac{\sin \pi z}{\pi z} \cdot \frac{\pi}{z \cos \pi z} = \infty$ ($\frac{\sin \pi z}{\pi z} \to 1$, $\frac{\pi}{z \cos \pi z} \to \infty$)

$\Rightarrow z = 0$ pol de ordinul 1 $\in$ Int γ

$\cos \pi z = 0 \Rightarrow \cos \pi z = \frac{e^{i\pi z} + e^{-i\pi z}}{2} = 0 \quad | \cdot e^{i\pi z} \Rightarrow$

$\Rightarrow e^{2i\pi z} + 1 = 0 \Rightarrow e^{2i\pi z} = -1 \Rightarrow$

$\Rightarrow 2\pi z i = \{ i(\pi + 2k\pi), k \in \mathbb{Z} \} \Rightarrow z = \{ \frac{1}{2} + k, k \in \mathbb{Z} \}$

$\notin$ Int γ

$$I = 2\pi i \operatorname{Rez}(f,0) = 2\pi i \lim_{z\to 0} z\cdot \frac{\operatorname{tg}\pi z}{z^2} =$$

$$= 2\pi i \lim_{z\to 0} \frac{\sin \pi z}{\pi z}\cdot \frac{\pi}{\cos \pi z} = 2\pi i \cdot \frac{\pi}{1} = 2\pi^2 i$$

f) $\displaystyle\int_{|z|=\pi} z^{-5}\cdot e^z\,dz = \int_{|z|=\pi} \frac{e^z}{z^5}\,dz \overset{\text{Not}}{=\!=} I$

$z=0$ pol de ordinul 5

$$I = 2\pi i \operatorname{Rez}(f,0)$$

$$\operatorname{Rez}(f,0) = \frac{1}{4!}\lim_{z\to 0}\left(z^5\cdot\frac{e^z}{z^5}\right)^{(4)} = \frac{1}{4!}\lim_{z\to 0} e^z = \frac{1}{4!}$$

$$I = 2\pi i\cdot\frac{1}{24} = \frac{\pi i}{12}$$

② Să se dezvolte funcția $f(z) = \dfrac{z}{z^2-1}$ pe domeniile:

a) $|z-1|<2$

$|z-1|<2 \Rightarrow \left|\frac{z-1}{2}\right|<1$ (avem nevoie să fie <1 pt a aplica formulele de la serii) $\Rightarrow$

$\Rightarrow$ formăm $\frac{z-1}{2}$ în funcție

$$f(z) = \frac{z}{z^2-1} = \frac{1}{2}\cdot\left(\frac{1}{z+1}+\frac{1}{z-1}\right) = \frac{1}{2}\left(\frac{1}{z-1+2}+\frac{1}{z-1}\right)$$

$$= \frac{1}{2}\left(\frac{1}{2}\,\frac{1}{\frac{z-1}{2}+1}+\frac{1}{z-1}\right) = \frac{1}{2}\left[\frac{1}{2}\sum_{n\geq 0}\underbrace{(-1)^n\cdot\left(\frac{z-1}{2}\right)^n}_{\left|\frac{z-1}{2}\right|<1}+\frac{1}{z-1}\right]$$

SAU $\displaystyle f(z) = \frac{z}{z^2-1} = \frac{z}{(z-1)(z+1)} = \frac{z}{z-1}\cdot\frac{1}{z-1+2}$

$$= \left(1+\frac{1}{z-1}\right)\cdot\frac{1}{z-1+2} = \left(1+\frac{1}{z-1}\right)\cdot\frac{1}{2}\cdot\frac{1}{\frac{z-1}{2}+1}$$

$$= \frac{1}{2}\left(1+\frac{1}{z-1}\right)\cdot\frac{1}{2}\cdot\sum_{n\geq 0}(-1)^n\cdot\left(\frac{z-1}{2}\right)^n$$

! $\frac{z-1}{2}$ e obligatoriu să fie în serie

b) $|z-1|>2$

$|z-1|>2 \Rightarrow 1>\frac{2}{|z-1|} \Rightarrow 1>\left|\frac{2}{z-1}\right|$ => formăm $\frac{2}{z-1}$ sau ceva mai mic

$$f(z)=\frac{z}{z^2-1}=\frac{1}{2}\cdot\left(\frac{1}{z+1}+\frac{1}{z-1}\right)=\frac{1}{2}\left(\frac{1}{z-1+2}+\frac{1}{z-1}\right)$$

$$=\frac{1}{2}\left(\frac{1}{z-1}\cdot\frac{1}{1+\frac{2}{z-1}}+\frac{1}{z-1}\right)=\frac{1}{2}\left(\frac{1}{z-1}\cdot\sum_{n\geq 0}(-1)^n\cdot\left(\frac{2}{z-1}\right)^n+\frac{1}{z-1}\right)$$

$$=\frac{1}{2}\frac{1}{z-1}\left(\sum_{n\geq 0}(-1)^n\cdot\left(\frac{2}{z-1}\right)^n+1\right)$$

$$\frac{1}{|z-1|}<\frac{2}{|z-1|}<1$$

3) Să se dezvolte funcția $f(z)=\frac{z^2+4z+1}{-z^3-2z^2+z+2}$ în jurul punctului $z_0=0$ și $z_0=-1$

	z^3	z^2	z	z^0
	-1	-2	1	2
1	-1	-3	-2	0
-1	-1	-2	0	

$$-z^3-2z^2+z+2=-(z-1)(z+1)(z+2)$$

$$\frac{z^2+4z+1}{-z^3-2z^2+z+2}=\frac{A}{z+2}+\frac{B}{z+1}+\frac{C}{z-1}$$

$$\Rightarrow A(z+1)(z-1)+B(z+2)(z-1)+C(z+2)(z+1)=z^2+4z+1$$

$$z^2(A+B+C)+z(B+3C)+(-A-2B+2C)=z^2+4z+1$$

$$\begin{cases} A+B+C=1 \\ B+3C=4 \\ -A-2B+2C=1 \end{cases} \Rightarrow \begin{cases} A+4-3C+C=1 \to A-2C=-3 \\ B=4-3C \\ -A-8+6C+2C=1 \to 8C-A=9 \end{cases}$$

$$6C=6 \Rightarrow C=1$$

$$A=2C-3=2-3=-1$$

$$B=4-3=1$$

$$f(z)=\frac{-1}{z+2}+\frac{1}{z+1}+\frac{1}{z-1}$$

pt. $z_0=0$ $\quad f(z)=\frac{1}{2}\cdot\frac{-1}{1+\frac{z}{2}}+\frac{1}{z+1}-\frac{1}{1-z}$

$$f(z)=-\frac{1}{2}\cdot\sum_{n\geq 0}(-1)^n\left(\frac{z}{2}\right)^n+\sum_{n\geq 0}(-1)^n z^n-\sum_{n\geq 0}z^n$$

$$f(z)=\sum_{n\geq 0}z^n\left((-1)^{n+1}\cdot\frac{1}{2^{n+1}}+(-1)^n-1\right) \qquad |z|<1$$

pt. $z_0=-1$ $\quad f(z)=\frac{-1}{z+1+1}+\frac{1}{z+1}+\frac{1}{z+1-2}$

$$f(z)=\frac{-1}{z+1+1}+\frac{1}{z+1}+\frac{1}{2}\cdot\frac{1}{\frac{z+1}{2}-1}$$

$$f(z)=-\sum_{n\geq 0}(-1)^n(z+1)^n+\frac{1}{z+1}-\frac{1}{2}\sum_{n\geq 0}\left(\frac{z+1}{2}\right)^n$$

$$f(z)=\frac{1}{z+1}+\sum_{n\geq 0}(z+1)^n\left((-1)^{n+1}-\frac{1}{2^{n+1}}\right) \qquad |z+1|<1$$

④ Calculați

a) $\int_{|z|=2} \frac{e^{iz}}{z(z^2+1)^2}\,dz \overset{\text{Not.}}{=\!=} I$

$z=0$ pol de ordinul 1 $\in Int|z|=2$
$z=\pm i$ pol de ordinul 2 $\in Int|z|=2$ $\xrightarrow{\text{Th. Rez.}}$

$$\Rightarrow I = 2\pi i\left(Rez(f,0)+Rez(f,i)+Rez(f,-i)\right)$$

$$Rez(f,0)=\lim_{z\to 0}\left[z\cdot\frac{e^{iz}}{z(z^2+1)^2}\right]=\frac{1}{1}=1$$

$$Rez(f,i)=\lim_{z\to i}\left[(z-i)^2\frac{e^{iz}}{z(z^2+1)^2}\right]^{1}$$

$$(z+i)^2$$

$$=\lim_{z\to i}\frac{e^{iz}\cdot i\,z(z+i)^2-e^{iz}\cdot\left[(z+i)^2+z\cdot 2(z+i)\right]}{z^2(z+i)^{4\,3}}=$$

$$=\lim_{z\to i}\frac{e^{iz}\left[iz^2-z-z-i-2z\right]}{z^2(z+i)^3}$$

$$=\frac{e^{i^2}\left(i^3-i-i-i-2i\right)}{i^2(2i)^3}=\frac{e^{-1}(-i-5i)}{(-1)\cdot 8\cdot(-1)\cdot i}=\frac{-6ie^{-1}}{8i}$$

$$Rez(f,i)=\frac{-3e^{-1}}{4}$$

$$\operatorname{Rez}(f,-i)=\lim_{z\to -i}\left[\cancel{(z+i)^2}\cdot\frac{e^{iz}}{z\cancel{(z^2+1)^2}}\right]^{1}$$
$$(z-i)^2$$

$$=\lim_{z\to -i}\frac{ie^{iz}z(z-i)^2-e^{iz}\left[(z-i)^2+z\cdot 2(z-i)\right]}{z^2(z-i)^4}$$

$$=\frac{ie(-i)(-2i)^2-e\left[(-2i)^2-2i(-2i)\right]}{(-i)^2(-2i)^4}=\frac{e\cdot 4\cdot(-1)-e[-4-4]}{(-1)2^4\cdot}$$

$$=\frac{-4e+8e}{-16}=\frac{4e}{-16}=\frac{-e}{4}$$

$$\int=2\pi i\left(1-\frac{3}{4e}-\frac{e}{4}\right)$$

b) $\displaystyle\int_{|z|=\sqrt{2}}\frac{z^3e^{1/z}}{z+1}dz \overset{\text{Not}}{=\!=} \int$

$z+1=0\Rightarrow z=-1$ pol de ordinul 1 $\in$ Int $|z|=\sqrt{2}$

$\lim_{z\to 0} e^{1/z}$ ∄ NU EXISTĂ pt. că

$z_n^1=\frac{1}{2n\pi i}$; $e^{\frac{1}{z_n^1}}=e^{2n\pi i}=\cos(2n\pi)+i\sin(2n\pi)=1$ $\Rightarrow$

$z_n^1\to 0$

$z_n^2=\frac{1}{n}$ $\quad e^{\frac{1}{n}}\to 1$

$z_n^2\to 0$

$\Rightarrow \alpha\neq 0$

$\Rightarrow z=0$ punct singular esențial $\in$ Int $|z|=\sqrt{2}$

$\xrightarrow[\text{Rezid}]{\text{Th.}}$ $I = 2\pi i \left(\text{Rez}(f,-1) + \text{Rez}(f,0) \right)$

$$\text{Rez}(f,-1) = \lim_{z \to -1} \left[(z+1) \cdot \frac{z^3 e^{1/z}}{z+1} \right] = (-1)^3 \cdot e^{-1} = -e^{-1}$$

$$\frac{z^3 e^{1/z}}{z+1} = z^3 \left(\sum_{n \geq 0} \frac{\left(\frac{1}{z}\right)^n}{n!} \right) \left(\sum_{n \geq 0} (-1)^n \cdot z^n \right)$$

$$= z^3 \left(1 + \frac{1}{z} + \frac{1}{2z^2} + \dots \right) \left(1 - z + z^2 - z^3 + \dots \right)$$

$$\text{Rez}(f,0) = \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \dots \Rightarrow \text{Rez}(f,0) = e^{-1} - \frac{1}{2} + \frac{1}{6} - \dots$$

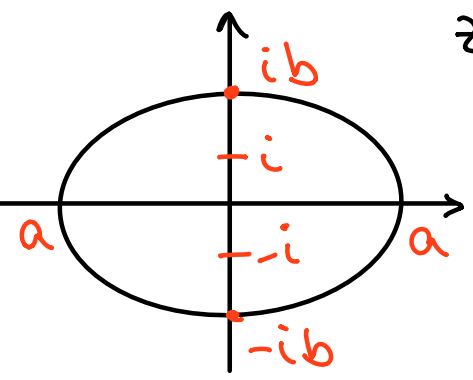
$$e^{-1} = \sum_{n \geq 0} \frac{(-1)^n}{n!} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} + \dots$$

$$I = 2\pi i \left(-e^{-1} + e^{-1} - \frac{1}{2} + \frac{1}{6} - \dots \right) = 2\pi i \left(-\frac{1}{2} + \frac{1}{6} - \dots \right)$$

c) $\displaystyle\int_{\gamma} \frac{\sin \frac{\pi}{z}}{1+z^2} dz \overset{\text{Not}}{=} I$ $\qquad \gamma: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad b > 1$

$1 + z^2 = 0 \Rightarrow z = \pm i$ poli de ordinul 1 $\in$ Int γ

$z = 0$ punct singular esențial $\in$ Int γ



$\xrightarrow[\text{Rezid.}]{\text{Th.}}$ $I = 2\pi i \left(\text{Rez}(f,i) + \text{Rez}(f,-i) + \text{Rez}(f,0) \right)$

$$\operatorname{Rez}(f,-i)=\lim_{z\to -i}\left[(z+i)\cdot\frac{\sin\frac{\pi}{z}}{1+z^2}\right]=\frac{\sin\frac{\pi}{-i}}{-2i}=\frac{\operatorname{sen}\pi i}{-2i}$$

$z-i$

$$=\frac{i\sin(\pi i)}{2}=\frac{i}{2}\cdot i\operatorname{sh}\pi=-\frac{\operatorname{sh}\pi}{2}$$

$$\operatorname{Rez}(f,i)=\lim_{z\to i}\left[(z-i)\cdot\frac{\sin\frac{\pi}{z}}{1+z^2}\right]=\frac{\sin\frac{\pi}{i}}{2i}=\frac{\sin(-\pi i)}{2i}$$

$z+i$

$$=\frac{-1}{2i}\cdot i\operatorname{sh}\pi=-\frac{\operatorname{sh}\pi}{2}$$

$$f(z)=\frac{\sin\frac{\pi}{z}}{1+z^2}=\sum_{n\geq 0}(-1)^n\cdot\frac{\left(\frac{\pi}{z}\right)^{2n+1}}{(2n+1)!}\cdot\sum_{n\geq 0}(-1)^n\cdot z^{2n}$$

$$=\left(\frac{\pi}{z}-\left(\frac{\pi}{z}\right)^3\cdot\frac{1}{3!}+\left(\frac{\pi}{z}\right)^5\cdot\frac{1}{5!}-\dots\right)\left(1-z^2+z^4-z^6+\dots\right)$$

$$\operatorname{Rez}(f,0)=\pi+\frac{\pi^3}{3!}+\frac{\pi^5}{5!}+\dots=\operatorname{sh}\pi$$

$$y=2\pi i\left(-\frac{\operatorname{sh}\pi}{2}-\frac{\operatorname{sh}\pi}{2}+\operatorname{sh}\pi\right)=0$$